Efficient Agent Training for Computer Use

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Paper version: arXiv 2505.13909v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Scaling up high-quality trajectory data has long been a critical bottleneck for developing human-like computer use agents.*

How few **human** trajectories suffice if a frontier model fills in alternative actions?

Motivation: three open gaps

1. **Scarce data.** High-quality trajectories are extremely scarce.
2. **One path.** A human records one solution; many actions are valid.
3. **Costly.** End-to-end Claude rollouts need VMs and online runs.

Evaluation setup

WindowsAgentArena-V2 · 141 | OSWorld · 369 tasks | Qwen2.5-VL-72B base | Claude 3.7 Sonnet teacher | 30-step budget | 1280 × 720 screenshots | runs / seeds not stated

- **All online** (VM success); no offline step accuracy.
- **Held out:** OSWorld (Linux), Windows-only training.

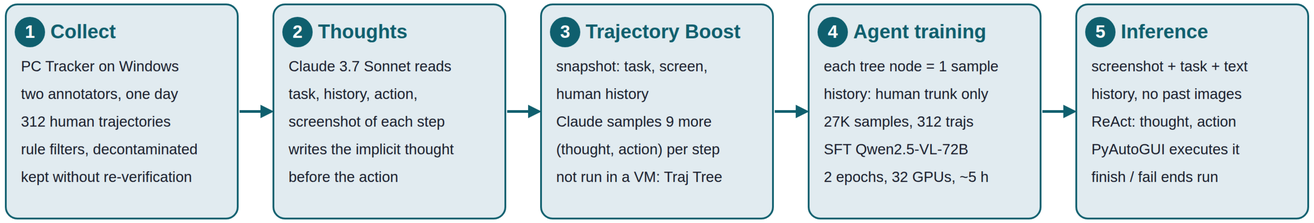
Our read (CUA Radar)

72	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out OS; strong baselines; matched distillation; cost; ablation; code.
- **Flags:** none recorded; variance unclear.
- Pattern C21. Authors' numbers, not rerun.

Method: human steps as snapshots, Claude adds alternative actions

- **Sees / acts:** screenshot, task and text history in; ReAct thought, then click, drag, type or hotkey via PyAutoGUI.
- **New:** Trajectory Boost samples 9 extra (thought, action) pairs per human step; none is executed.
- **Signal:** plain SFT on 27K samples; no RL, no verifier.

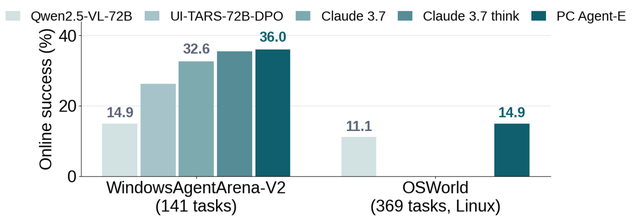


Our redraw (not the paper's figure): the pipeline and the per-step Traj Tree.

$$s = \frac{\#\,\text{actions used in training}}{\#\,\text{actions in human trajectory}} = 9 + 1 = 10$$

data scaling factor: s = 1 is human data alone

Key results: online success rate (%)

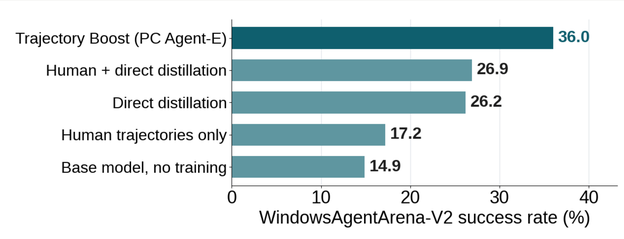


Tables 3 and 4 redrawn: single values, no std reported.

36.0 vs 32.6 (Claude 3.7 Sonnet) and 14.9 (Qwen2.5-VL-72B)

SO WHAT → *WindowsAgentArena-V2: 141 tasks, infeasible tasks removed by the authors.*

Ablation: WindowsAgentArena-V2 success (%)



§5.3 and Table 7, redrawn: all trained from Qwen2.5-VL-72B.

SO WHAT → *Claude-sampled actions on human states beat human-only data and end-to-end distillation. Distillation took about 900 h of collection against 3 h. The paper's 50-trajectory review finds the gain is in planning, not knowledge or grounding.*

Headline Numbers

36.0%
WAA-V2 success rate
base Qwen2.5-VL-72B: 14.9%; Claude 3.7: 32.6%

- **141%** rel. gain (stated)
- **17.2%** human data only
- **34%** rel. gain, OSWorld

Controls (WAA-V2 unless noted)

CONTROL	PC AGENT-E	COMPARED WITH
Direct distillation	36.0	26.2
Human + direct distillation	36.0	26.9
Steps 15 / 30 / 50	26.9 / 36.0 / 31.4	11.3 / 14.9 / 11.9 base
OSWorld feas. / infeas.	10.9 / 63.3	4.4 / 86.7 base

Takeaway

312 human trajectories plus 9 Claude-sampled actions per step lift Qwen2.5-VL-72B from 14.9 to 36.0, through better planning.

27K samples, 312 trajs | 3 h vs about 900 h | 7B model: 5.0 to 6.4

Scan to Read

Paper | Code